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1. Which line was causing the RecursionError, and why does storing into self._temperature fix it?

The RecursionError was caused by line 24, self.temperature = value, because it called the temperature setter repeatedly. Changing it to self._temperature = value stores the value directly in the private variable and stops the recursion.

2. Bug 5 still lets 99 through even after fixing the setter. What was the constructor doing wrong?

The constructor was directly storing the temperature on line 11 using self._temperature = temperature. This bypassed the validation in the temperature setter. Changing it to self.temperature = temperature makes the constructor use the setter and reject invalid temperatures.

3. Why should is_energy_saving be calculated each time instead of stored once

is_energy_saving should be calculated each time because the temperature can change. If it is stored once in the constructor on line 14, it can become outdated. Calculating self.temperature >= 25 each time ensures the result always reflects the current temperature.